

NavalForge Concept

Preliminary conceptual design report

12 m rescue and support craft — NF-DEMO-RESCUE-12M — Revision P1

Engineering gate: CONDITIONAL — preliminary results with active engineering reservations

Technical notice: Preliminary engineering result. It does not replace the responsible engineer, formal approval, validated software, detailed structural analysis or verified input data.

Main indicators

Indicator	Value	Unit
Adherence	100.0	%
Displacement	14569.26	kg
Maximum speed	35.00	kn
Required installed power	382.64	kW
Range	557.56	nmi
Corrected GM	0.710	m
Freeboard	0.977	m

Requirement matrix

ID	Requirement	Actual	Result
REQ12-LOA	Overall length shall not exceed 12.50 m	12.0 m	PASS
REQ12-SPEED	Maximum speed shall be at least 32 kn	35.0 kn	PASS
REQ12-RANGE	Range shall be at least 220 nmi	557.5582257430633 nmi	PASS
REQ12-GM	Preliminary corrected GM shall exceed 0.50 m	0.709702491315202 m	PASS
REQ12-FB	Preliminary freeboard shall exceed 0.60 m	0.9772466208572445 m	PASS
REQ12-PAYLOAD	Payload shall be at least 1000 kg	1000.0 kg	PASS

Warnings and assumptions

- GZ curve uses a wall-sided preliminary estimate; validate with inclined geometry before statutory use.
- Partially filled tanks present; free-surface correction applied.
- Consumption must be replaced by verified engine/propulsor curves for contractual use.
- Speed is below the preferred planing-method range; displacement effects are simplified.
- All project and price data are synthetic demonstrations.
- Composite laminate properties are placeholders for conceptual screening.
- Conceptual parametric hull used unless verified offsets are supplied.
- Demonstrative prices are synthetic and are not current quotations.

Selected alternatives

NF-ECO — NF-DEMO-RESCUE-12M-V001

Lowest demonstrative cost among variants passing every mandatory requirement.

NF-BALANCED — NF-DEMO-RESCUE-12M-V009

Highest auditable multicriteria balance of cost, performance, range, GM and risk.

NF-PERFORMANCE — NF-DEMO-RESCUE-12M-V003

Highest combined performance and technical-margin screening among compliant variants.